

§4.2

P1.  $A = \begin{pmatrix} 1 & 1 \\ -2 & -2 \end{pmatrix}$

$$P(\lambda) = \det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & 1 \\ -2 & -2-\lambda \end{pmatrix}$$

$$= \lambda(\lambda+1) = 0$$

$$\Rightarrow \lambda_1 = 0, \lambda_2 = -1$$

stable.

P2.  $A = \begin{pmatrix} -3 & -4 \\ 2 & 1 \end{pmatrix}$

$$P(\lambda) = \det(A - \lambda I) = \det \begin{pmatrix} -3-\lambda & -4 \\ 2 & 1-\lambda \end{pmatrix}$$

$$= \lambda^2 + 2\lambda + 5 = 0$$

$$\Rightarrow \lambda = -1 \pm 2i$$

Asymptotically stable

P3.  $A = \begin{pmatrix} -5 & 3 \\ -1 & 1 \end{pmatrix}$

$$P(\lambda) = \det(A - \lambda I) = \det \begin{pmatrix} -5-\lambda & 3 \\ -1 & 1-\lambda \end{pmatrix}$$

$$= \lambda^2 + 4\lambda - 2 = 0$$

$$\Rightarrow \lambda_1 = -2 + \sqrt{6}, \lambda_2 = -2 - \sqrt{6}$$

$\lambda_1 > 0 \Rightarrow$  unstable.

P5.  $A = \begin{pmatrix} -7 & 1 & -6 \\ 10 & -4 & 12 \\ 2 & -1 & 1 \end{pmatrix}$

$$P(\lambda) = \det(A - \lambda I) = \det \begin{pmatrix} -7-\lambda & 1 & -6 \\ 10 & -4-\lambda & 12 \\ 2 & -1 & 1-\lambda \end{pmatrix}$$

$$= -\lambda^3 - 10\lambda^2 - 31\lambda - 30$$

$$= -(\lambda+2)(\lambda+3)(\lambda+5)$$

$$\Rightarrow \lambda_1 = -2, \lambda_2 = -3, \lambda_3 = -5$$

$\Rightarrow$  Asymptotically stable.

P6.  $A = \begin{pmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{pmatrix}$

$$P(\lambda) = \det(A - \lambda I) = \det \begin{pmatrix} 3-\lambda & 2 & 4 \\ 2 & -\lambda & 2 \\ 4 & 2 & 3-\lambda \end{pmatrix}$$

$$= -(\lambda+1)^2(\lambda-8) = 0$$

$$\Rightarrow \lambda_1 = -1, (\text{double root})$$

$$\lambda_2 = 8 > 0$$

unstable.

P16. Solution:

We know that the stability of any solution of  $\dot{x} = \begin{pmatrix} -1 & -1 \\ 2 & -1 \end{pmatrix} x + \begin{pmatrix} 1 \\ 5 \end{pmatrix}$  is equivalent to the stability of the equilibrium solution  $x \equiv 0$  of  $\dot{x} = \begin{pmatrix} -1 & -1 \\ 2 & -1 \end{pmatrix} x$ .

$$A = \begin{pmatrix} -1 & -1 \\ 2 & -1 \end{pmatrix}$$

$$P(\lambda) = \det(A - \lambda I) = \det \begin{pmatrix} -1-\lambda & -1 \\ 2 & -1-\lambda \end{pmatrix}$$

$$= \lambda^2 + 2\lambda + 3 = 0$$

$$\Rightarrow \lambda = -1 \pm \sqrt{2}i$$

$\Rightarrow$  Asymptotically stable.